

① Two numbers are 15% and 20% less than a third number. What percentage is the first number as a percentage of second number.

- A) 105% B) 106.25%
C) 107.5% D) 116.5%

② In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party? (GATE - 18) —

- (a) 2 : 3 (b) 1 : 1
(c) 3 : 2 (d) 2 : 1

- 3 One percent of the people country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft? (GATE -14) — 2M
- (a) 3.0 (b) 2.5
(c) 1.5 (d) 1.25

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of the election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 6 votes then what was the total number of voters? (GATE – 2011, 16)

- A) 300 B) 200
C) 600 D) 800

✓ A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?

(EC/ME – 2016) (2M)

- | | |
|--------|--------|
| (a) 15 | (b) 33 |
| (c) 35 | (d) 37 |

On time $\rightarrow x\%$

up to 15 minutes delayed $\rightarrow 43\%$

15 - 30 minutes delayed $\rightarrow 17\%$

30 - 60 minutes delayed $\rightarrow 12\%$

More than 60 minutes delayed $\rightarrow 3\%$

The chart above describes from a certain airport on a certain day. If 1,200 flights were delayed, how many flights departed on time?

- (a) 350 (b) 300
(c) 250 (d) 400

In a competitive examination in State A, 6% candidates got selected from the total appeared candidates. State B had an equal number of candidates appeared and 7% candidates got selected with 80 more candidates got selected than A. What was the number of candidates appeared from each State?

- (a) 7600 (b) 8000
(c) 8400 (d) Data inadequate

→ In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners, only 294 million access the internet. Among these users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country? (GATE - 2019) — 1M

(a) 10.50 (b) 14.70

(c) 15.00 (d) 50.00

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✓ In an examination Suresh scored 22% of the maximum marks and failed the exam by 52 marks, whereas Ramesh scored 45% of the maximum marks and got 40 marks more than the pass mark. Then find the maximum marks of the exam?

(a) 300 (b) 200
(c) 400 (d) 500

$$1T + 1S = 95$$

10% fall / 20% rise
90% T 120% S

✓ One kg of tea and one kg of sugar together cost Rs. 95. If the price of tea fall by 10% and that of sugar rises by 20%, then the price of one kg of each combined comes to Rs. 90. The original price of tea in Rs. per kg is_____.

- (a) Rs. 72 (b) Rs. 55
(c) Rs. 60 (d) Rs. 80

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10% of the voters did not cast their vote in an election between two candidates. 10% of the votes polled were found invalid. The successful candidate got 54% of the valid votes and won by a majority of 1620 votes. The number of votes enrolled on the voters list was:

- (a) 25000 (b) 33000
(c) 35000 (d) 40000

% CHANGE & % COMPARISON

A number is mistakenly divided by 5 instead of multiplied by 5; find the percentage change in the result due to the mistake.

- A) 96% B) 480%
C) 2400% D) 100%

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A saves 20% of his income. His income is increased by 20% and so he increased his expenditure by 30%. What is the percentage change in his savings?

- a) 20% fall
- b) 4% fall
- c) 20% rise
- d) 4% rise

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GATE-19: The radius as well as the height of a circular cone increases by 10%. The percentage increase in its volume is _____.

- A) 17.1 B) 21.0
C) 33.1 D) 72.8

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Fiscal deficit was 4% of the GDP in 2015 and that increased to 5% in 2016. If the GDP increased by 10% from 2015 to 2016, the percentage increase in the actual fiscal deficit is____ (GATE - 2019) — 21

- (a) 37.50 (b) 25.00
(c) 35.70 (d) 10.00

Population of state X increased by $x\%$ and the population of state Y increased by $y\%$ from 2001 to 2011. Assume that x is greater than y . Let P be the ratio of the population of state X to state Y in a given year. The percentage increase in P from 2001 to 2011 is _____.

(GATE - 2019)

- (a) $100(x - y)/(100 + y)$
- (b) $x - y$
- (c) x/y
- (d) $100(x - y)/(100 + x)$

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In an examination, A got 10% marks less than B, B got 25% more than C, and C got 20% less than D. If A got 360 out of 500, the percentage marks obtained by D was:

- | | |
|-------|-------|
| A) 70 | B) 75 |
| C) 80 | D) 85 |